



Distributed gradient descent: when the model does not fit

Part 2 of 2

Carlos Cotrini



Section 1

The memory ledger

1

Four workers, one average

Eight examples, one gradient number each

+2

-2

+3

-1

+4

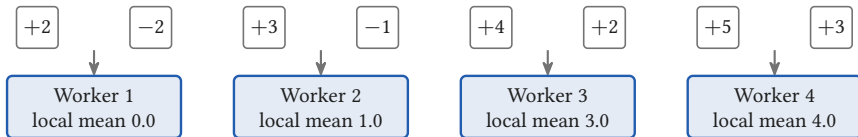
+2

+5

+3

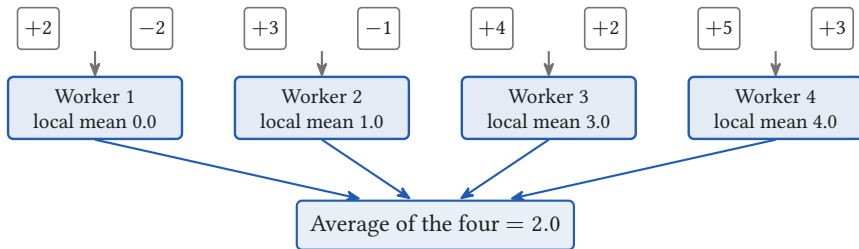
Four workers, one average

Eight examples, one gradient number each



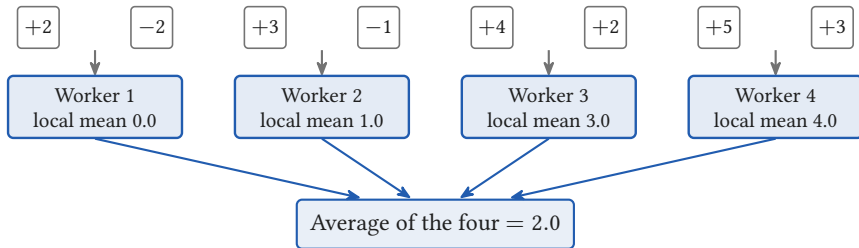
Four workers, one average

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Four workers, one average

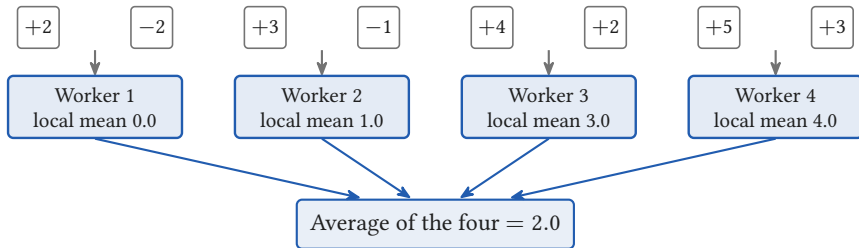
Eight examples, one gradient number each



One all-reduce averages them, and it is exactly the single machine gradient

Four workers, one average

Eight examples, one gradient number each

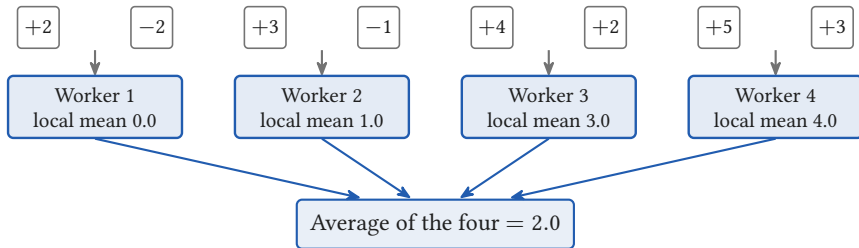


One all-reduce averages them, and it is exactly the single machine gradient

- The gradient of a mean is the mean of the gradients

Four workers, one average

Eight examples, one gradient number each



One all-reduce averages them, and it is exactly the single machine gradient

- The gradient of a mean is the mean of the gradients
- Every worker held a whole copy of the model. Today it does not fit.

Sixteen bytes per parameter

Mixed precision Adam, one parameter

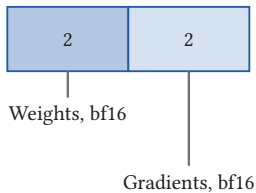


Weights, bf16

[ZeRO](#) (Rajbhandari et al., SC20) Section 3.1, verbatim: $2\Psi + 2\Psi + K\Psi = 16\Psi$ bytes, with Ψ the parameter count and $K = 12$ the optimiser multiplier, a different K from part 1's gradient buffer.

Sixteen bytes per parameter

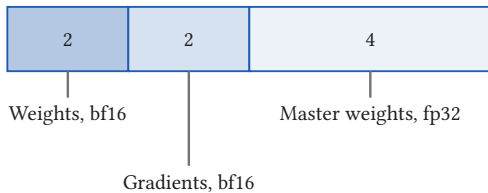
Mixed precision Adam, one parameter



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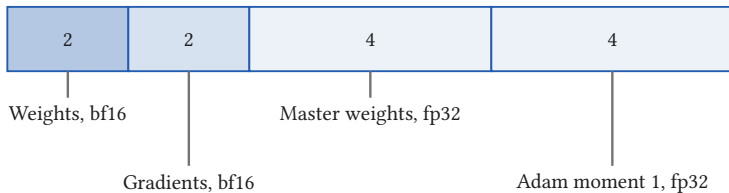
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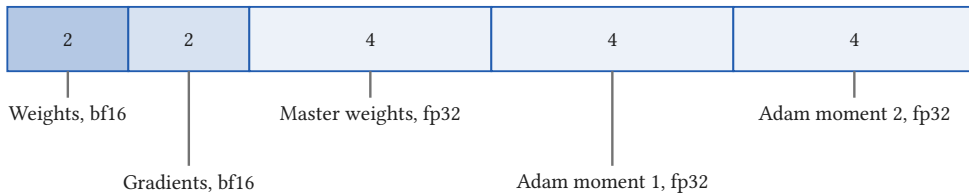
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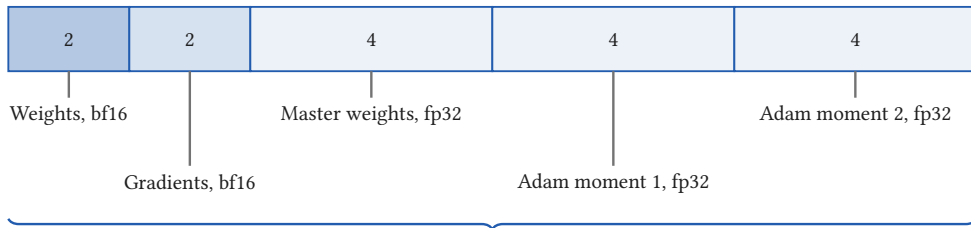
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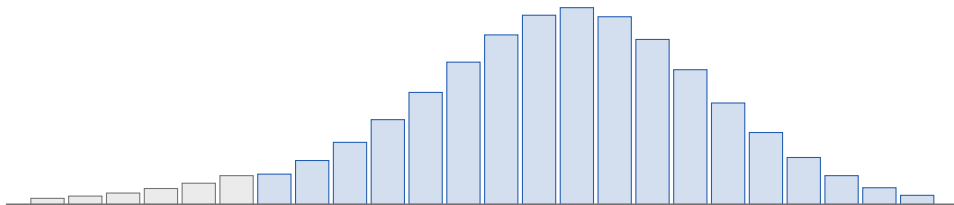
Mixed precision Adam, one parameter



$$2 + 2 + 4 + 4 + 4 = 16 \text{ bytes per parameter}$$

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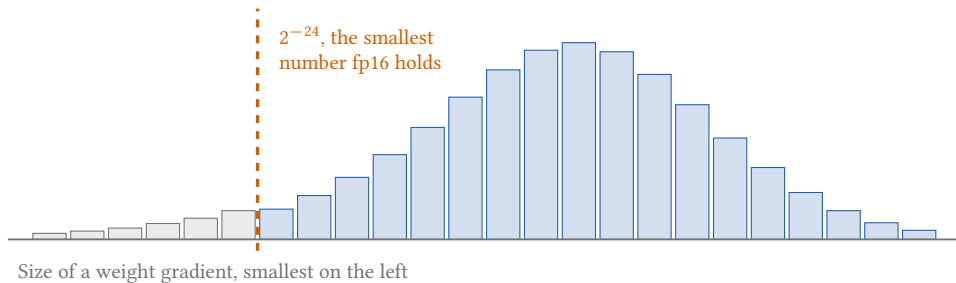
Why four of those bytes are fp32



Size of a weight gradient, smallest on the left

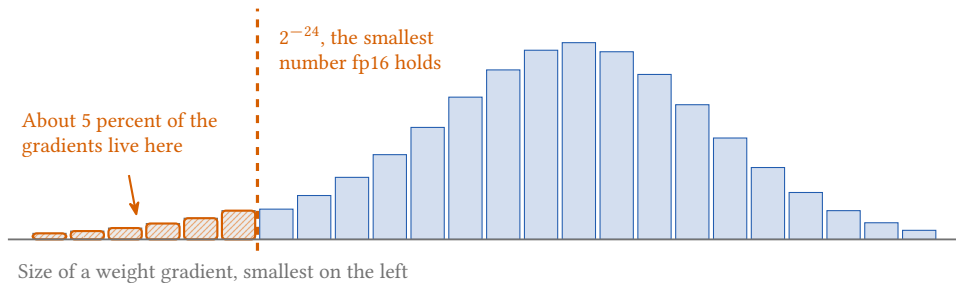
Schematic shape. The measured fact is the 5 percent, from [Micikevicius et al. 2018](#), Figure 2b.

Why four of those bytes are fp32



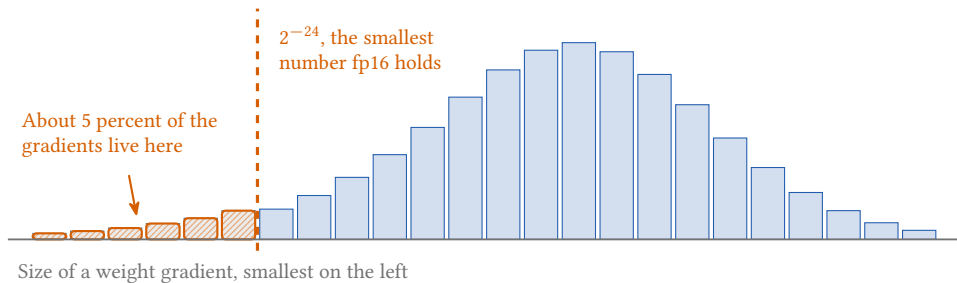
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Why four of those bytes are fp32



Schematic shape. The measured fact is the 5 percent, from [Micikevicius et al. 2018](#), Figure 2b.

Why four of those bytes are fp32



In fp16 they become exactly zero, and never recover.
That is why a master copy of the weights is kept in fp32.

Schematic shape. The measured fact is the 5 percent, from [Micikevicius et al. 2018](#), Figure 2b.

The ledger changes with the optimiser

Bytes per parameter

Pure bf16, Kahan summation



12

[ZeRO Section 3.1](#); [Ultra-Scale Playbook](#); [CS336 Lecture 7](#). [AdEMAMix](#) keeps three optimiser states, not two.

The ledger changes with the optimiser

Bytes per parameter



[ZeRO Section 3.1](#); [Ultra-Scale Playbook](#); [CS336 Lecture 7](#). [AdEMAMix](#) keeps three optimiser states, not two.

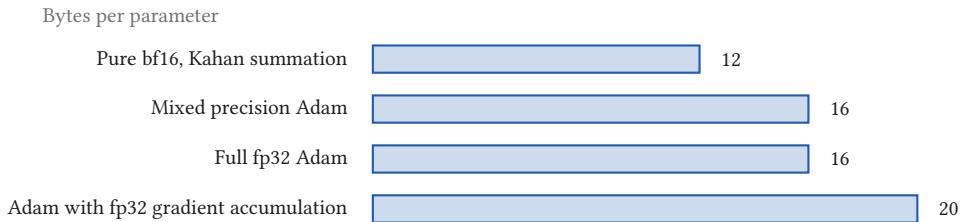
The ledger changes with the optimiser

Bytes per parameter



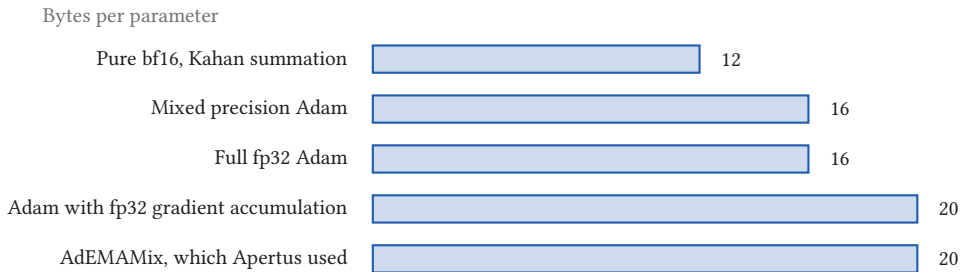
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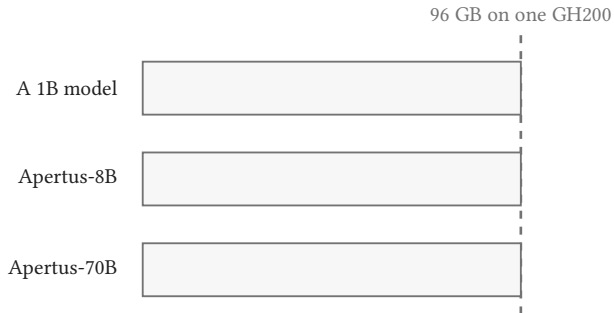
Bytes per parameter



Mixed precision does not reduce the model state. It redistributes it:
full fp32 Adam costs the same 16 bytes.

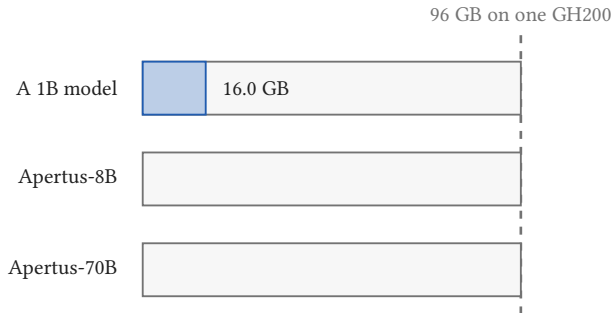
[ZeRO Section 3.1](#); [Ultra-Scale Playbook](#); [CS336 Lecture 7](#). AdEMAMix keeps three optimiser states, not two.

Three models, one accelerator



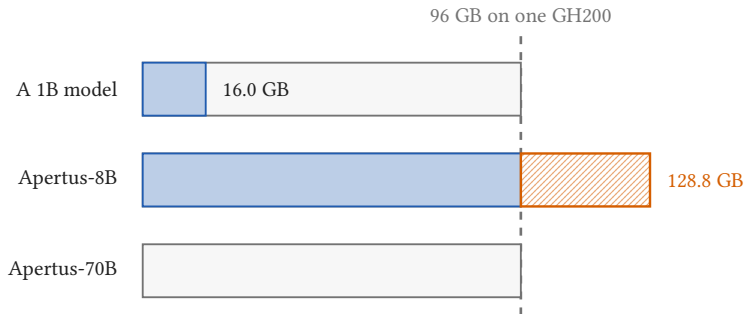
On an 80 GB card the wall is at 5.00×10^9 parameters. With Apertus's own optimiser, at 4.80×10^9 .

Three models, one accelerator



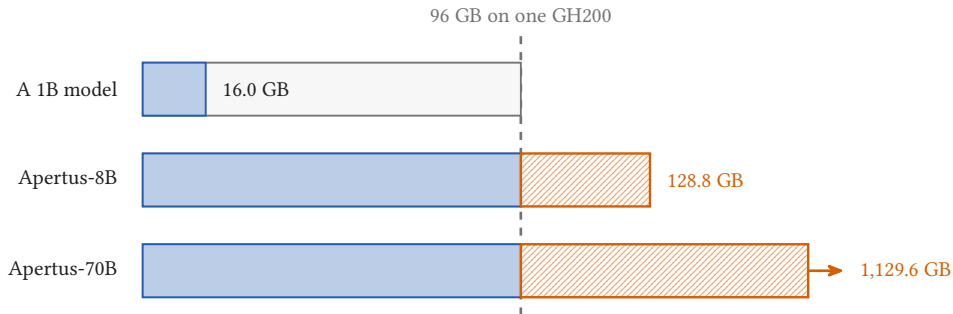
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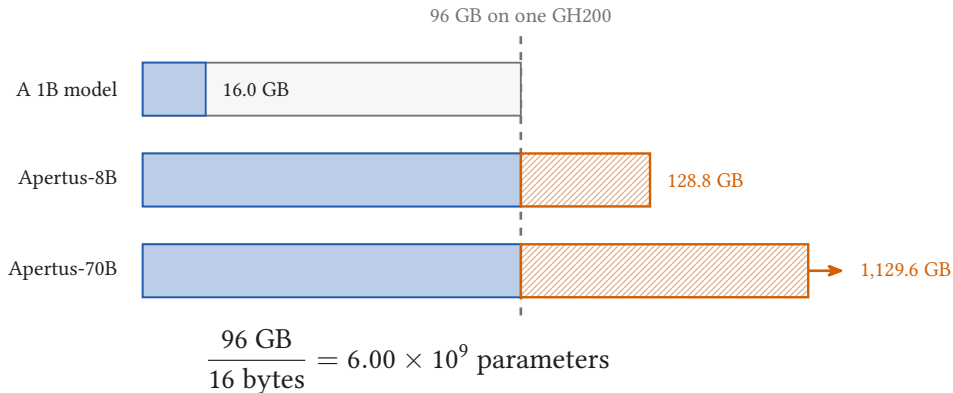
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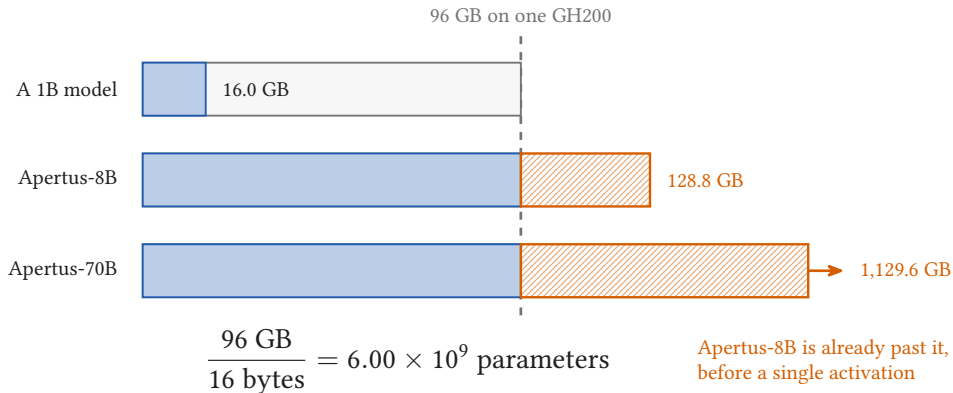
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Three models, one accelerator



On an 80 GB card the wall is at 5.00×10^9 parameters. With Apertus's own optimiser, at 4.80×10^9 .

What the ledger leaves out

One accelerator, 96 GB



What the ledger leaves out

One accelerator, 96 GB



- **Activations:** usually the larger number at long context

What the ledger leaves out

One accelerator, 96 GB



- **Activations:** usually the larger number at long context
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What the ledger leaves out

One accelerator, 96 GB



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- **Temporary buffers:** a flattened fp32 all-reduce buffer for a 1.5B model is 6 GB

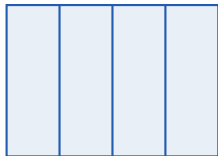
What the ledger leaves out

One accelerator, 96 GB



- **Activations:** usually the larger number at long context
- **CUDA context:** 1 to 2 GB on every GPU, before your model
- **Temporary buffers:** a flattened fp32 all-reduce buffer for a 1.5B model is 6 GB
- **Fragmentation:** ZeRO observed failures with over 30 percent nominally free

Three ways to cut it



Cut the matrix
tensor parallel

Three ways to cut it

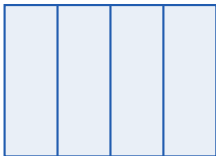


Cut the matrix
tensor parallel



Cut the depth
pipeline parallel

Three ways to cut it



Cut the matrix
tensor parallel

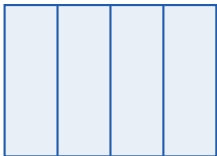


Cut the depth
pipeline parallel



Cut the state
ZeRO and FSDP

Three ways to cut it



Cut the matrix
tensor parallel



Cut the depth
pipeline parallel



Cut the state
ZeRO and FSDP

- And every large published run uses all three at once

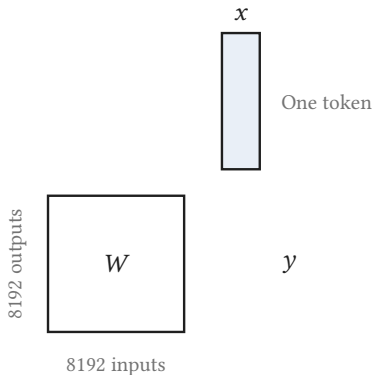


Section 2

Cut the matrix

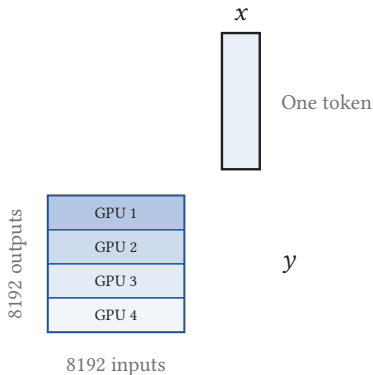
2

Cut the rows: each device owns some outputs



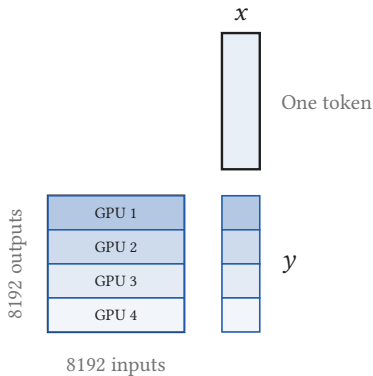
- Every device is given the whole of x

Cut the rows: each device owns some outputs



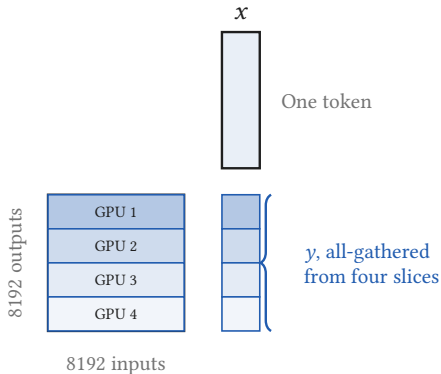
- Every device is given the whole of x
- Each owns 2048 of the 8192 outputs

Cut the rows: each device owns some outputs



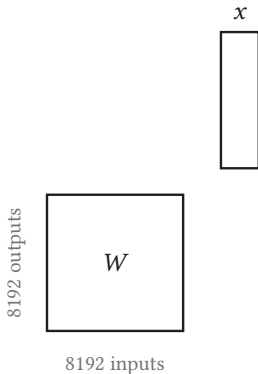
- Every device is given the whole of x
- Each owns 2048 of the 8192 outputs
- Each computes its own slice of y , alone

Cut the rows: each device owns some outputs



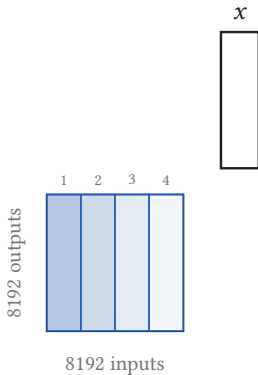
- Every device is given the whole of x
- Each owns 2048 of the 8192 outputs
- Each computes its own slice of y , alone
- The slices are concatenated, never summed

Cut the columns: each device owns some inputs



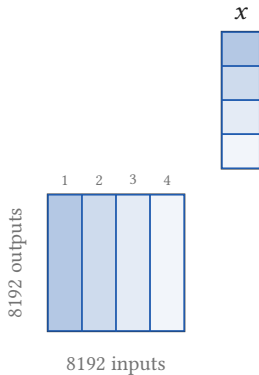
- Every device is given a slice of x

Cut the columns: each device owns some inputs



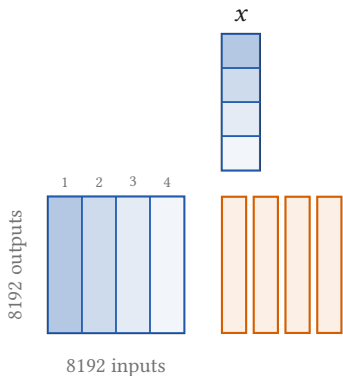
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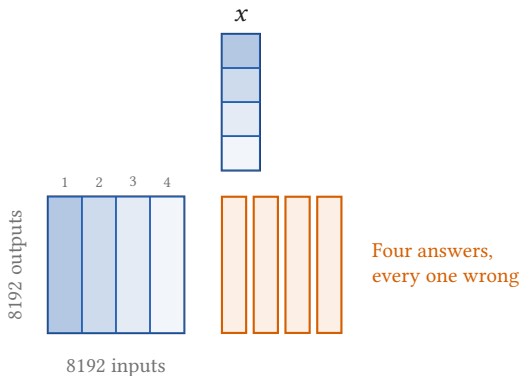
- Every device is given a slice of x
- Each owns 2048 of the 8192 inputs
- Each produces a full length answer

Cut the columns: each device owns some inputs



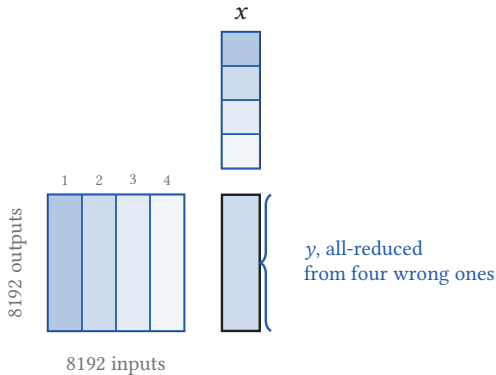
- Every device is given a slice of x
- Each owns 2048 of the 8192 inputs
- Each produces a full length answer
- Right shape, wrong value, on every device

Cut the columns: each device owns some inputs



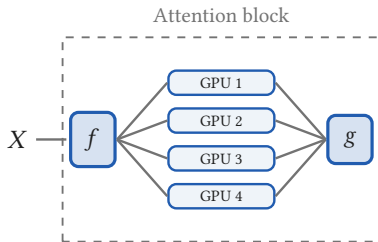
- Every device is given a slice of x
- Each owns 2048 of the 8192 inputs
- Each produces a full length answer
- Right shape, wrong value, on every device
- Only their sum is the layer's output

Cut the columns: each device owns some inputs



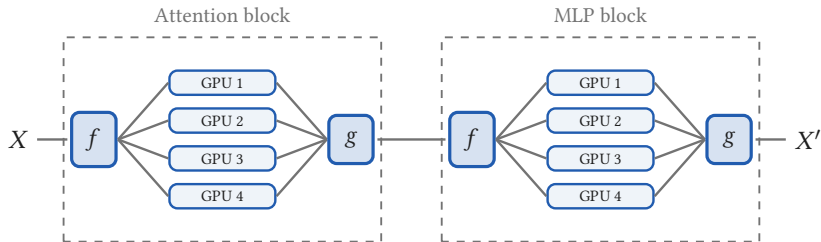
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Four collectives per layer



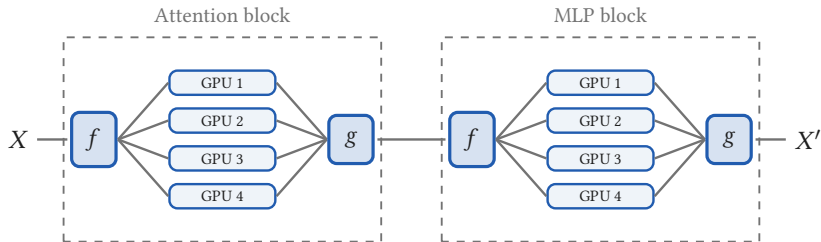
Megatron-LM (Shoeybi et al. 2019), Section 3 and Figure 4.

Four collectives per layer



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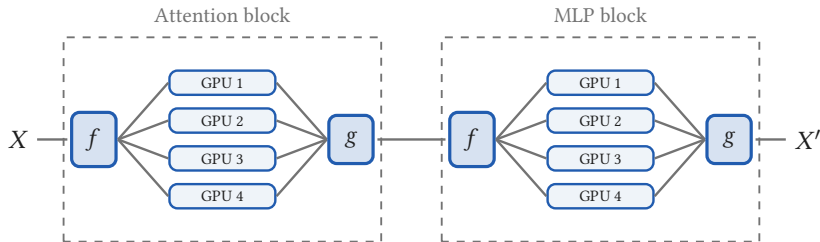
Four collectives per layer



f does nothing going forward, and all-reduces coming back

Megatron-LM (Shoeybi et al. 2019), Section 3 and Figure 4.

Four collectives per layer

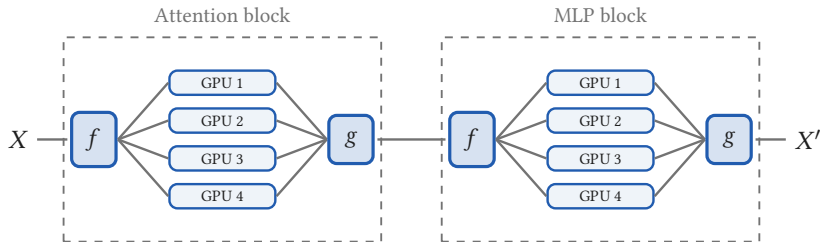


f does nothing going forward, and all-reduces coming back

g all-reduces going forward, and does nothing coming back

Megatron-LM (Shoeybi et al. 2019), Section 3 and Figure 4.

Four collectives per layer



f does nothing going forward, and all-reduces coming back

g all-reduces going forward, and does nothing coming back

Two all-reduces forward,
two backward, per layer

Megatron-LM (Shoeybi et al. 2019), Section 3 and Figure 4.

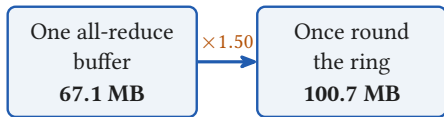
One buffer, four collectives, eighty layers

One all-reduce
buffer
67.1 MB

Apertus-70B: bf16, sequence 4096, micro batch 1, $d_{\text{model}} = 8192$, tensor parallel degree 4.

A second published expression for this volume disagrees, and disagrees in direction. Quote the collective count, not a volume.

One buffer, four collectives, eighty layers

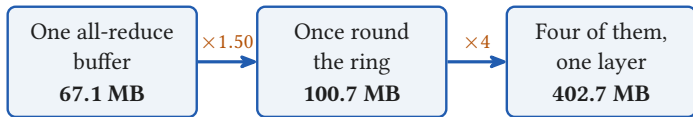


$2(P - 1)/P$ with $P = 4$, the tensor parallel degree

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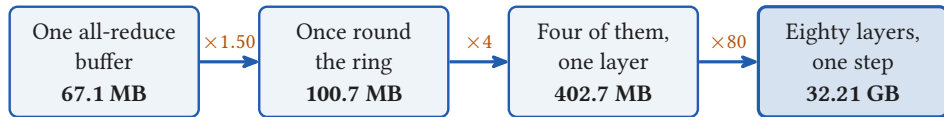


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One buffer, four collectives, eighty layers



$2(P - 1)/P$ with $P = 4$, the tensor parallel degree

And that is per step, on top of the data parallel all-reduce from yesterday.

Apertus-70B: bf16, sequence 4096, micro batch 1, $d_{\text{model}} = 8192$, tensor parallel degree 4.

A second published expression for this volume disagrees, and disagrees in direction. Quote the collective count, not a volume.

Why tensor parallelism stays inside the node

$$\frac{TP - 1}{2h} \times \frac{\text{peak FLOP/s}}{\text{peak bandwidth}}$$

Why tensor parallelism stays inside the node

$$\underbrace{\frac{TP - 1}{2h}}_{\text{how much must move}} \times \frac{\text{peak FLOP/s}}{\text{peak bandwidth}}$$

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$$\underbrace{\frac{TP - 1}{2h}}_{\text{how much must move}} \times \underbrace{\frac{\text{peak FLOP/s}}{\text{peak bandwidth}}}_{\text{how fast the link is}} \leq 1$$

Why tensor parallelism stays inside the node

$$\underbrace{\frac{TP - 1}{2h}}_{\text{how much must move}} \times \underbrace{\frac{\text{peak FLOP/s}}{\text{peak bandwidth}}}_{\text{how fast the link is}} \leq 1$$

900 GB/s

Inside one node 

0.201, so it overlaps

Why tensor parallelism stays inside the node

$$\underbrace{\frac{TP - 1}{2h}}_{\text{how much must move}} \times \underbrace{\frac{\text{peak FLOP/s}}{\text{peak bandwidth}}}_{\text{how fast the link is}} \leq 1$$

900 GB/s

Inside one node  0.201, so it overlaps

25 GB/s per GPU

Across the fabric  7.251, so it does not

Why tensor parallelism stays inside the node

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900 GB/s

Inside one node  0.201, so it overlaps

25 GB/s per GPU

Across the fabric  7.251, so it does not

- Pinned inside the node in every large run: PTD-P at 8, Llama 3 405B at 8, Apertus-70B at 4

Why tensor parallelism stays inside the node

$$\underbrace{\frac{TP - 1}{2h}}_{\text{how much must move}} \times \underbrace{\frac{\text{peak FLOP/s}}{\text{peak bandwidth}}}_{\text{how fast the link is}} \leq 1$$

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Inside one node  0.201, so it overlaps

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Across the fabric  7.251, so it does not

- Pinned inside the node in every large run: PTD-P at 8, Llama 3 405B at 8, Apertus-70B at 4
- Capped by the key and value heads as well, and Apertus has 8



Section 3

Cut the depth, and the bubble

3

Eighty layers over eight devices

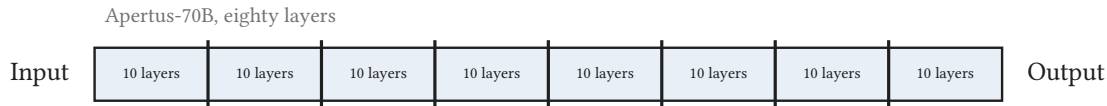
Apertus-70B, eighty layers

Input

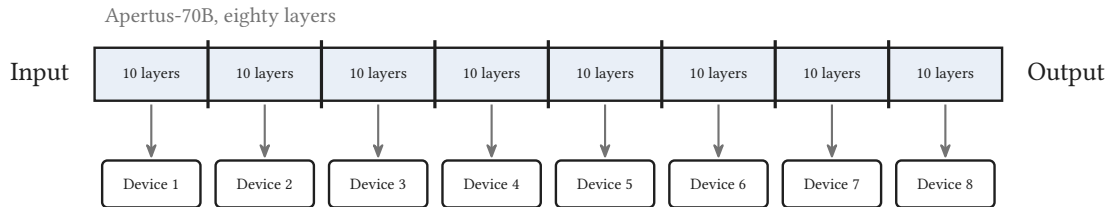


Output

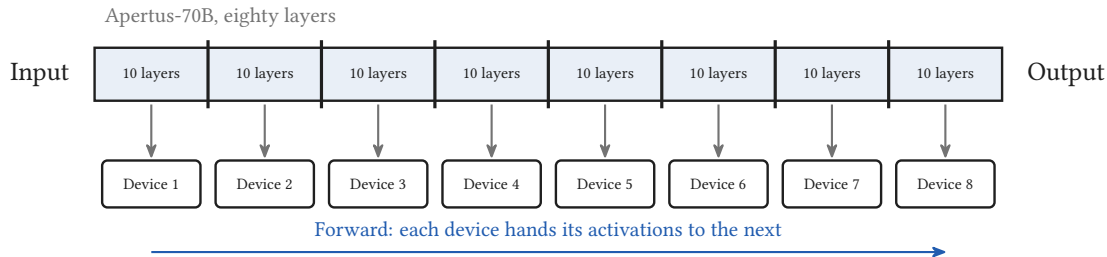
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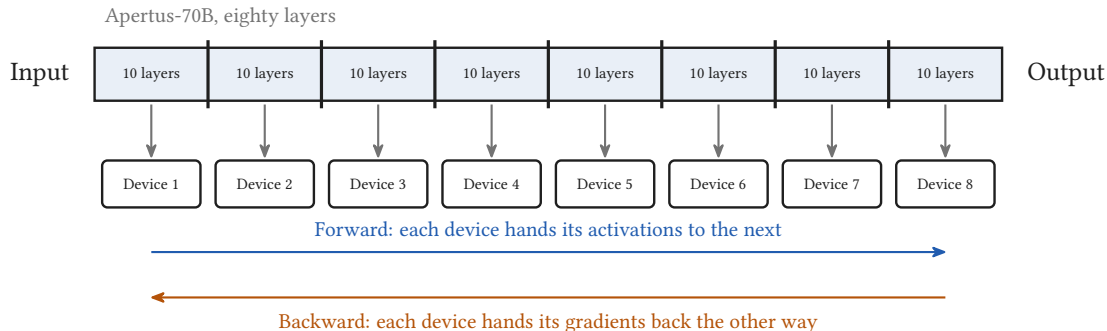
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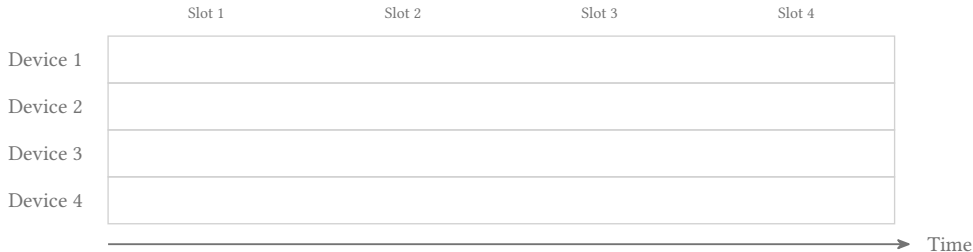
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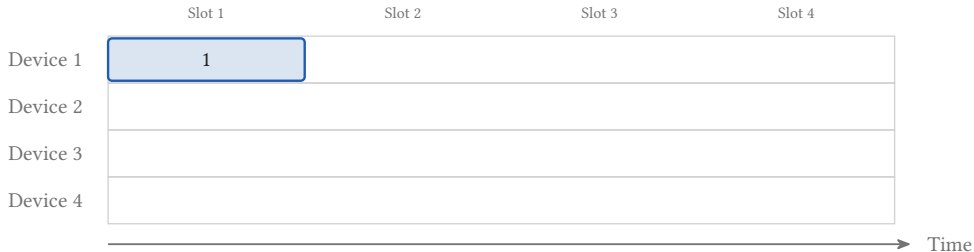
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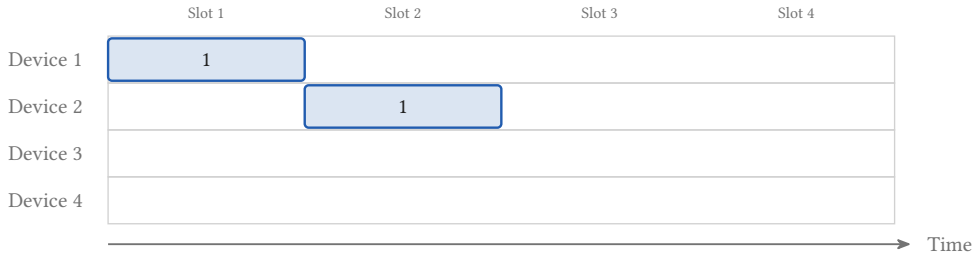
Four stages, one micro-batch



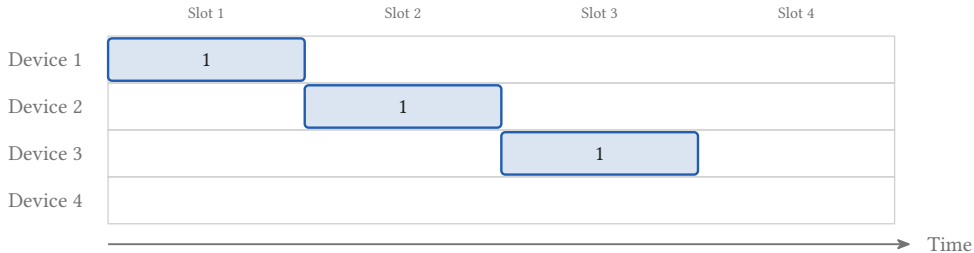
Four stages, one micro-batch



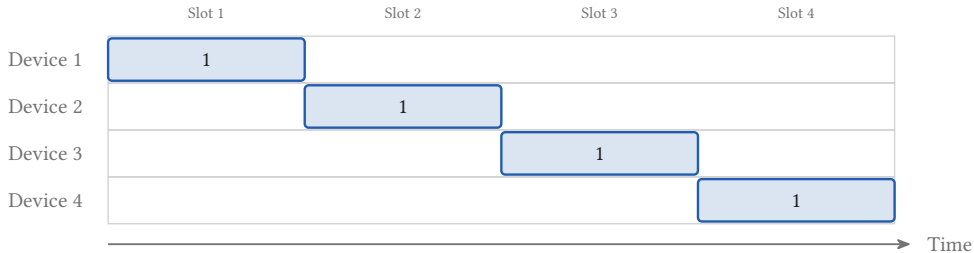
Four stages, one micro-batch



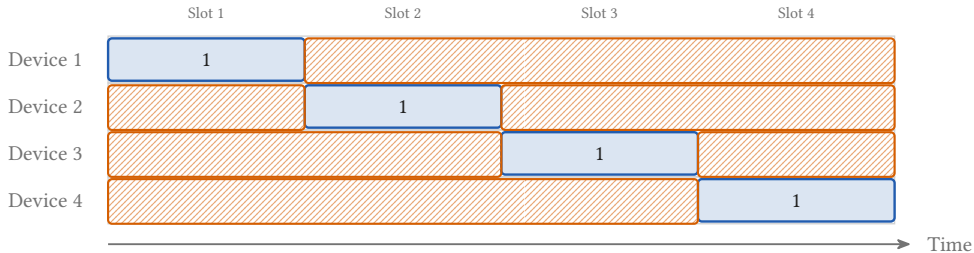
Four stages, one micro-batch



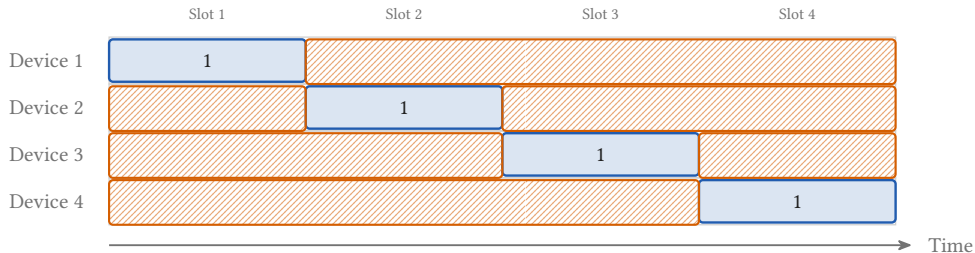
Four stages, one micro-batch



Four stages, one micro-batch



Four stages, one micro-batch



Three slots idle out of four: **75.0 percent** of the machine

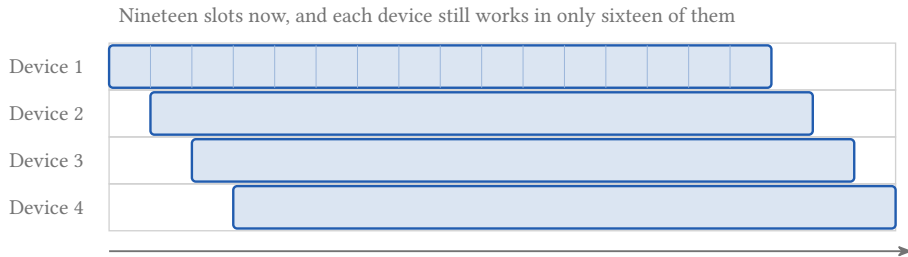
Four stages, sixteen micro-batches

Nineteen slots now, and each device still works in only sixteen of them



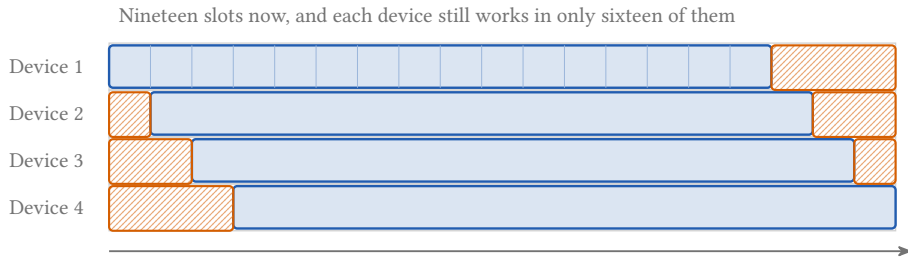
Each block is one micro-batch passing through one stage. The idle slots per device did not change: the run got longer.

Four stages, sixteen micro-batches



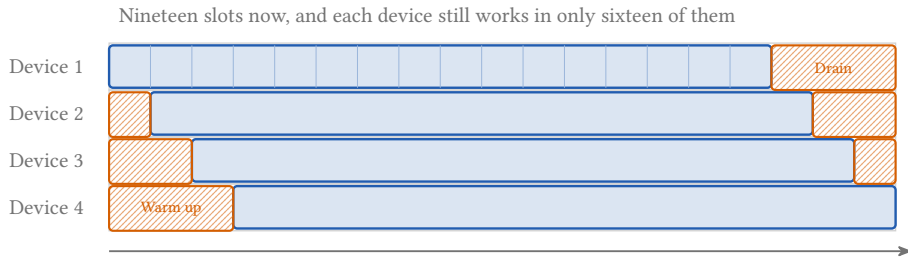
Each block is one micro-batch passing through one stage. The idle slots per device did not change: the run got longer.

Four stages, sixteen micro-batches



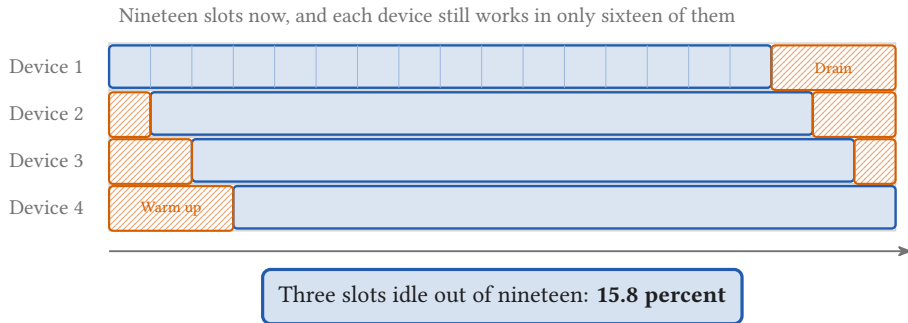
Each block is one micro-batch passing through one stage. The idle slots per device did not change: the run got longer.

Four stages, sixteen micro-batches



Each block is one micro-batch passing through one stage. The idle slots per device did not change: the run got longer.

Four stages, sixteen micro-batches



Each block is one micro-batch passing through one stage. The idle slots per device did not change: the run got longer.

The bubble fraction



P stages in the chain



M micro-batches in one batch

$$\text{Idle share} = \frac{P - 1}{M + P - 1}$$

GPipe (Huang et al. 2019). Assumes balanced stages, free hand-offs, equal micro-batches and a flush per batch.

The bubble fraction



P stages in the chain



M micro-batches in one batch

$$\text{Idle share} = \frac{\overbrace{\text{slots a device waits}}^{P-1}}{M + P - 1}$$

The bubble fraction



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The bubble fraction



P stages in the chain



M micro-batches in one batch

$$\text{Idle share} = \frac{\overbrace{\text{slots a device waits}}^{P-1}}{\underbrace{M + P - 1}_{\text{slots in the run}}}$$

Four stages, one micro-batch
 $P = 4, M = 1$



75.0 percent idle

The bubble fraction



P stages in the chain



M micro-batches in one batch

$$\text{Idle share} = \frac{\overbrace{\text{slots a device waits}}^{P-1}}{\underbrace{M + P - 1}_{\text{slots in the run}}}$$

Four stages, one micro-batch

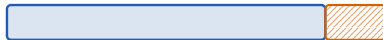
$$P = 4, M = 1$$



75.0 percent idle

Four stages, sixteen micro-batches

$$P = 4, M = 16$$



15.8 percent idle

The bubble fraction



P stages in the chain



M micro-batches in one batch

$$\text{Idle share} = \frac{\overbrace{\text{slots a device waits}}^{P-1}}{\underbrace{M + P - 1}_{\text{slots in the run}}}$$

Four stages, one micro-batch

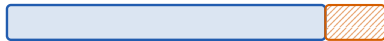
$$P = 4, M = 1$$



75.0 percent idle

Four stages, sixteen micro-batches

$$P = 4, M = 16$$



15.8 percent idle

A second convention divides by ideal compute time instead, and reports these same two pipelines as 300.0 and 18.8 percent. Never mix the two.

GPipe (Huang et al. 2019). Assumes balanced stages, free hand-offs, equal micro-batches and a flush per batch.

More stages makes it worse

	Micro-batches M							
Percent idle	1	2	4	8	16	32	64	256

More stages makes it worse

		Micro-batches M						
Percent idle	1	2	4	8	16	32	64	256
$P = 4$	75.0	60.0	42.9	27.3	15.8	8.6		

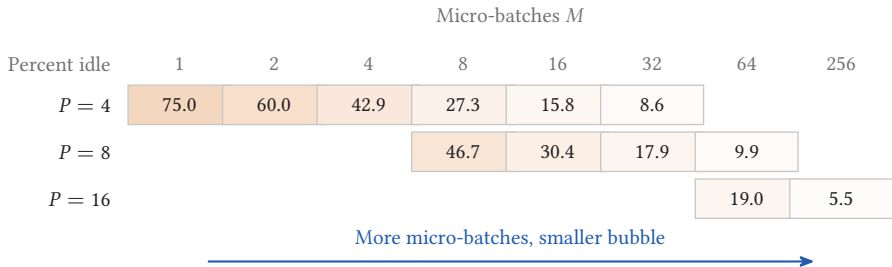
More stages makes it worse

	Micro-batches M							
Percent idle	1	2	4	8	16	32	64	256
$P = 4$	75.0	60.0	42.9	27.3	15.8	8.6		
$P = 8$				46.7	30.4	17.9	9.9	

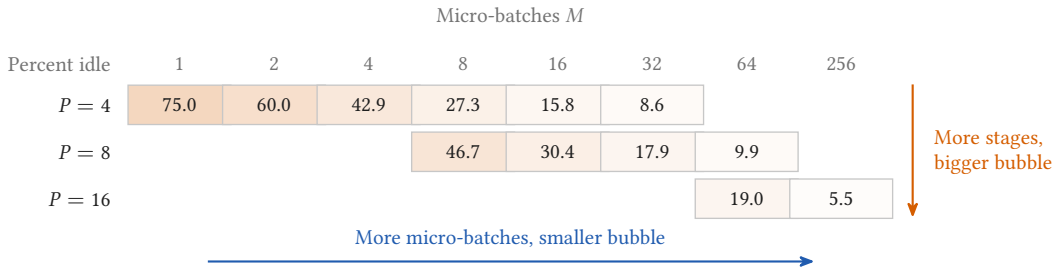
More stages makes it worse

	Micro-batches M							
Percent idle	1	2	4	8	16	32	64	256
$P = 4$	75.0	60.0	42.9	27.3	15.8	8.6		
$P = 8$				46.7	30.4	17.9	9.9	
$P = 16$							19.0	5.5

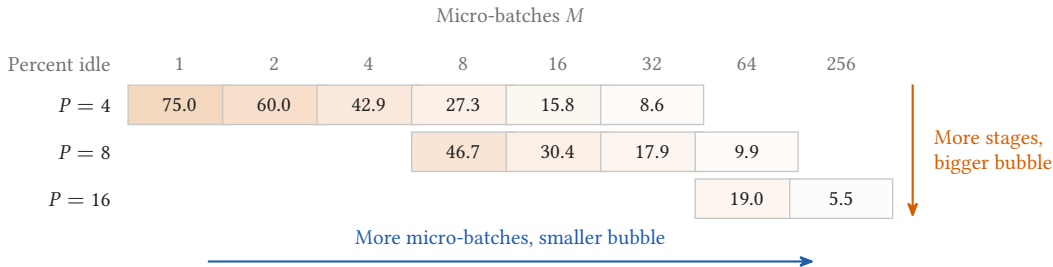
More stages makes it worse



More stages makes it worse

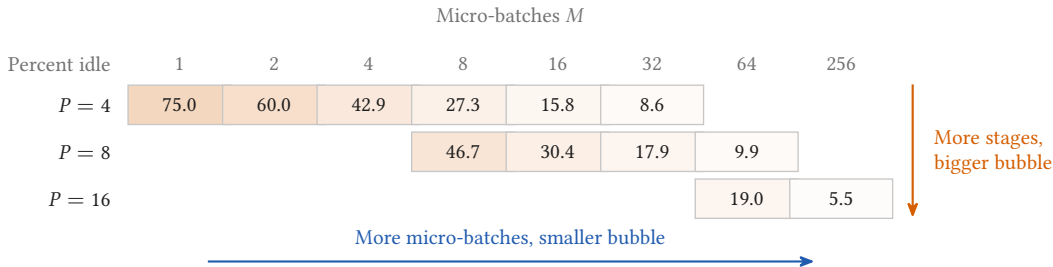


More stages makes it worse



- And M is capped: the micro-batches have to add up to the batch size you wanted

More stages makes it worse



- And M is capped: the micro-batches have to add up to the batch size you wanted
- One forward one backward does not shrink the bubble. It lowers peak activation memory from M micro-batches to P , which lets you raise M , and that is what shrinks it.



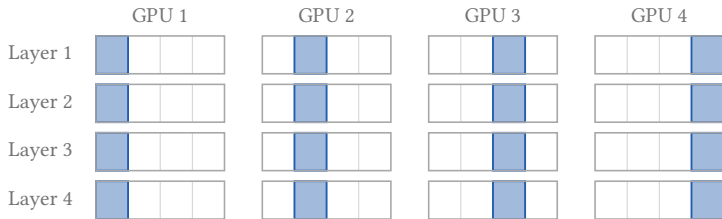
Section 4

Cut the state

4

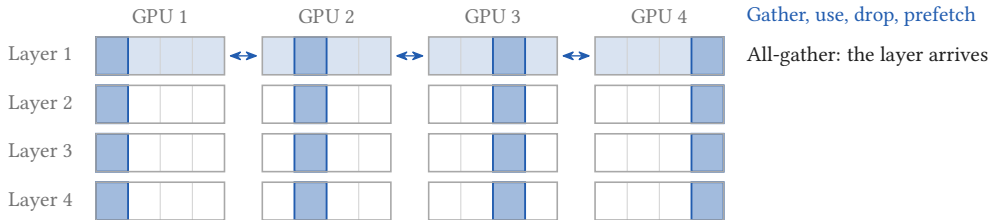
Nobody stores what they are not using

Each outlined box is one whole layer, marked off in quarters. The darker quarter is the only one this GPU owns.



Nobody stores what they are not using

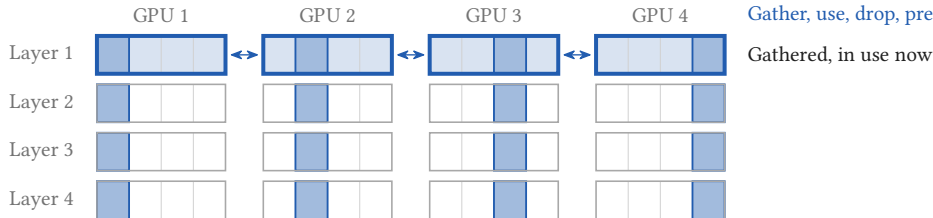
Each outlined box is one whole layer, marked off in quarters. The darker quarter is the only one this GPU owns.



Nobody stores what they are not using

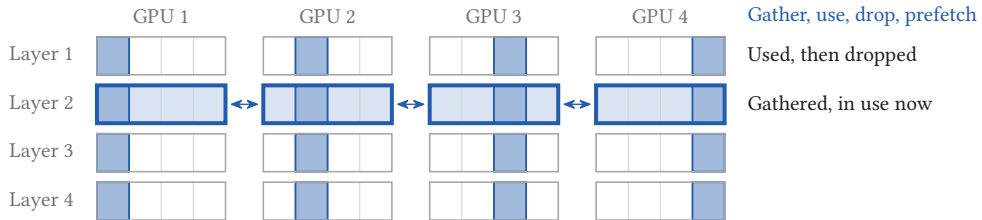
Each outlined box is one whole layer, marked off in quarters. The darker quarter is the only one this GPU owns.

Gather, use, drop, prefetch



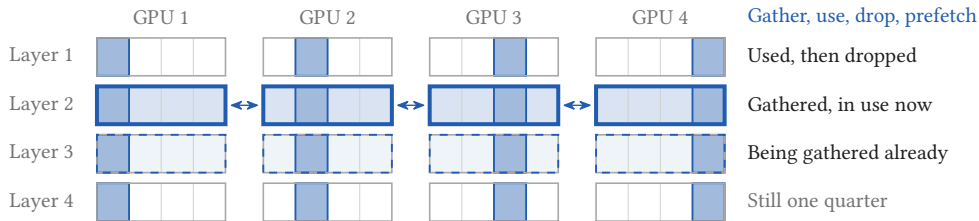
Nobody stores what they are not using

Each outlined box is one whole layer, marked off in quarters. The darker quarter is the only one this GPU owns.



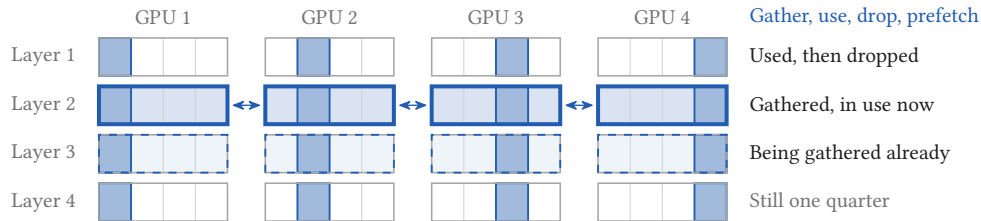
Nobody stores what they are not using

Each outlined box is one whole layer, marked off in quarters. The darker quarter is the only one this GPU owns.

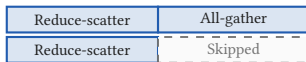


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Each outlined box is one whole layer, marked off in quarters. The darker quarter is the only one this GPU owns.



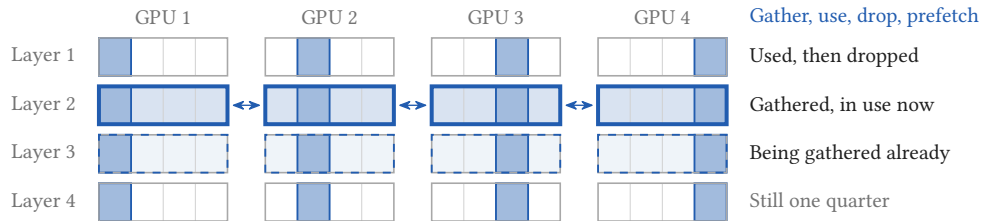
Part 1, gradient all-reduce
ZeRO-3 and FSDP



Each device updates only its own shard

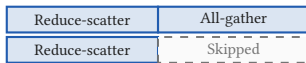
Nobody stores what they are not using

Each outlined box is one whole layer, marked off in quarters. The darker quarter is the only one this GPU owns.



Part 1, gradient all-reduce

ZeRO-3 and FSDP

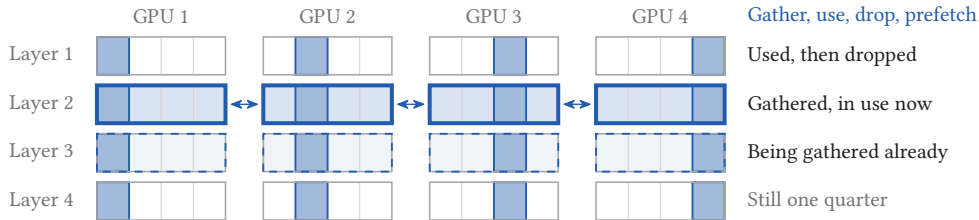


Each device updates only its own shard

- Every device does the same arithmetic on the same batch as before. Only residency changed.

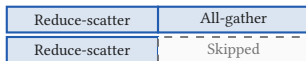
Nobody stores what they are not using

Each outlined box is one whole layer, marked off in quarters. The darker quarter is the only one this GPU owns. **Gather, use, drop, prefetch**



Part 1, gradient all-reduce

ZeRO-3 and FSDP

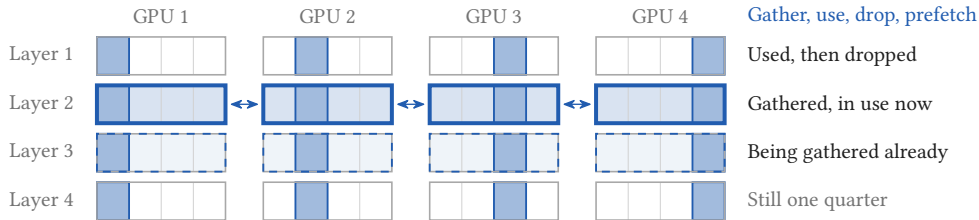


Each device updates only its own shard

- Every device does the same arithmetic on the same batch as before. Only residency changed.
- Peak residency is the layer in use, plus the one arriving

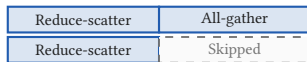
Nobody stores what they are not using

Each outlined box is one whole layer, marked off in quarters. The darker quarter is the only one this GPU owns. **Gather, use, drop, prefetch**



Part 1, gradient all-reduce

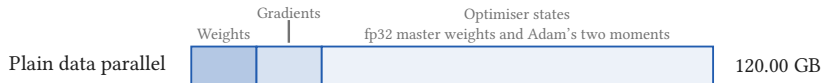
ZeRO-3 and FSDP



Each device updates only its own shard

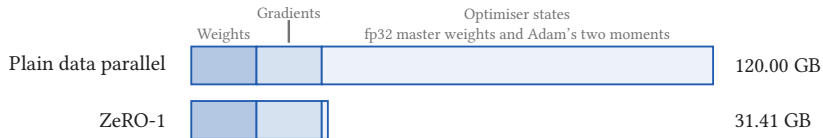
- Every device does the same arithmetic on the same batch as before. Only residency changed.
- Peak residency is the layer in use, plus the one arriving
- No device holds the whole model now, so part 1's gradient all-gather goes away

Three stages, one bar



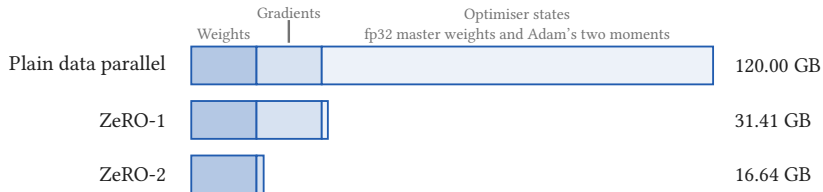
$\Psi = 7.5 \times 10^9$ parameters, 64 data parallel workers, mixed precision Adam. Traffic in elements, not bytes.

Three stages, one bar



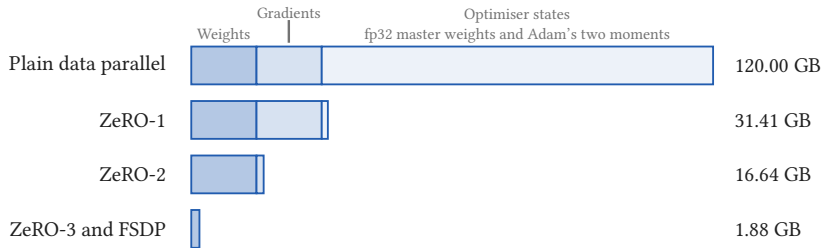
$\Psi = 7.5 \times 10^9$ parameters, 64 data parallel workers, mixed precision Adam. Traffic in elements, not bytes.

Three stages, one bar



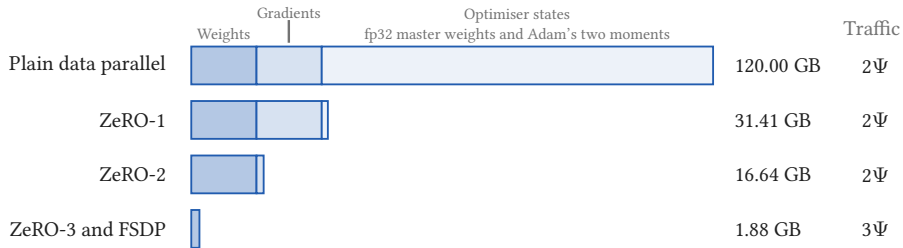
$\Psi = 7.5 \times 10^9$ parameters, 64 data parallel workers, mixed precision Adam. Traffic in elements, not bytes.

Three stages, one bar



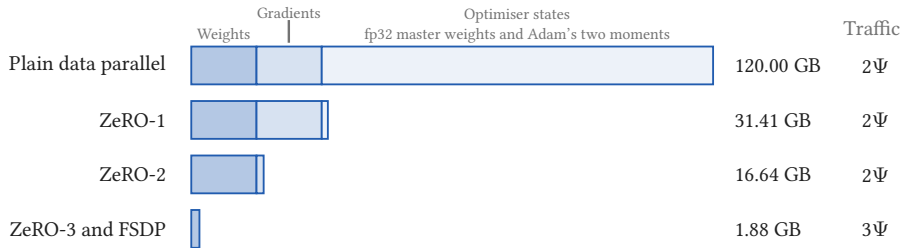
$\Psi = 7.5 \times 10^9$ parameters, 64 data parallel workers, mixed precision Adam. Traffic in elements, not bytes.

Three stages, one bar



$\Psi = 7.5 \times 10^9$ parameters, 64 data parallel workers, mixed precision Adam. Traffic in elements, not bytes.

Three stages, one bar



The headline 4x and 8x for the first two stages are limits as the worker count grows.

At 64 workers the realised factors are 3.82x and 7.21x: what is not sharded does not go away.

$\Psi = 7.5 \times 10^9$ parameters, 64 data parallel workers, mixed precision Adam. Traffic in elements, not bytes.

The same eight GPUs, eight times the model

Largest model that fits on one node of eight 80 GB accelerators

Plain data parallel

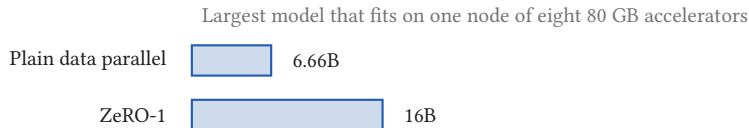


6.66B

CS336 Lecture 7, pure bf16 with fp32 master weights, so 12 bytes per parameter here rather than 16.

ZeRO-3 also adds two collectives per layer, which is why the rule of thumb keeps the data parallel degree under about 512.

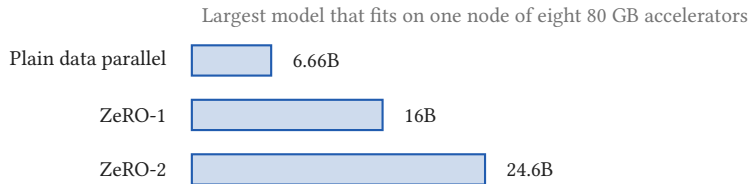
The same eight GPUs, eight times the model



CS336 Lecture 7, pure bf16 with fp32 master weights, so 12 bytes per parameter here rather than 16.

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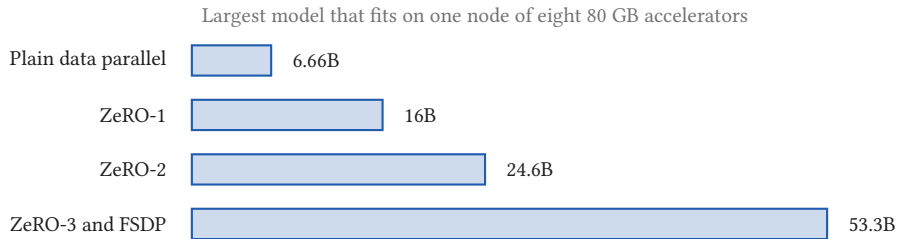
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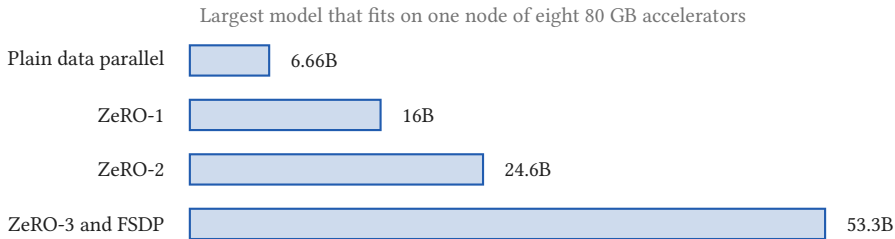
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CS336 Lecture 7, pure bf16 with fp32 master weights, so 12 bytes per parameter here rather than 16.

ZeRO-3 also adds two collectives per layer, which is why the rule of thumb keeps the data parallel degree under about 512.

The same eight GPUs, eight times the model



No change to the model, the data or the hardware. One configuration flag.

CS336 Lecture 7, pure bf16 with fp32 master weights, so 12 bytes per parameter here rather than 16.

ZeRO-3 also adds two collectives per layer, which is why the rule of thumb keeps the data parallel degree under about 512.



Section 5

All three at once

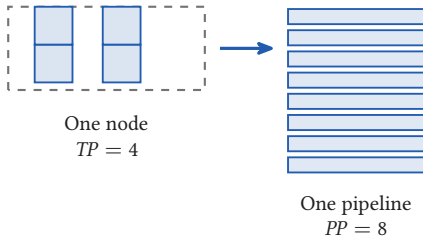
5

Four times eight times sixty-four

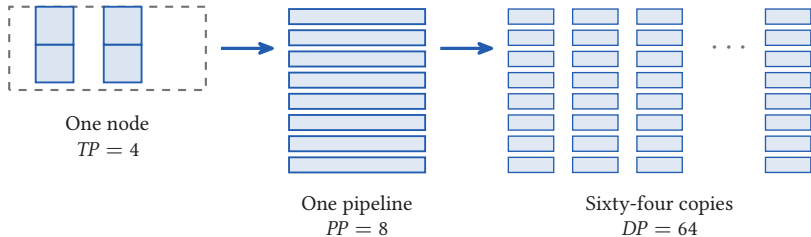


One node
 $TP = 4$

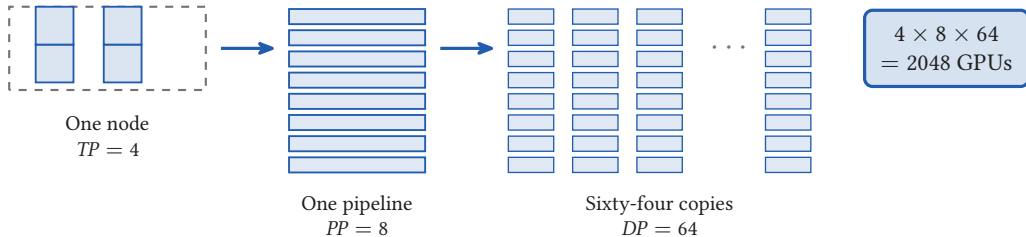
Four times eight times sixty-four



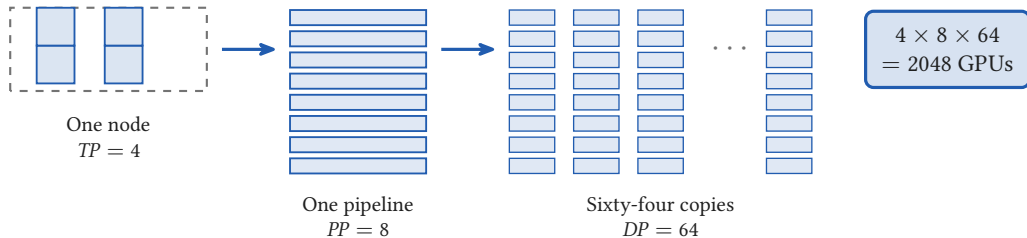
Four times eight times sixty-four



Four times eight times sixty-four



Four times eight times sixty-four



Tensor parallel inside the node, pipeline parallel across nodes, data parallel across the copies.